
Can This Agent Even Do That?

A Decidable Goal-Achievability Type Discipline for LLM-Synthesized Agent Skills

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Abstract

1 Modern autonomous agents rely on natural-language skills that specify what tools
2 they can use, how they should interact, and what outcome they should achieve.
3 Such skills can be written by human experts or generated with the help of LLMs.
4 Yet a basic question is often left to runtime: **can the goal described by a skill
5 actually be achieved?** A skill may require a capability that is unavailable, impose
6 a constraint that no available action can satisfy, or describe a multi-agent interaction
7 that can never complete. These failures can waste substantial computation before
8 they are discovered. We present a *skill compiler* that answers this question before
9 execution without relying on an LLM judge. The compiler combines capability-
10 based reachability from automated planning with communication checking from
11 **simple multiparty sessions (SMPS)**, directly checking a goal-marked global proto-
12 col together with the agent’s available capabilities and initial state [1, 2]. An LLM
13 may compact the natural-language skill into a structured formal pack containing
14 capabilities, preconditions, effects, goals, and the global protocol; the resulting
15 pack is then checked deterministically. The checker is backed by a Coq-verified ab-
16 straction theorem: when it reports *impossible*, the goal is indeed impossible under
17 the formal pack it received. An *achievable* result is deliberately weaker and may
18 still require further checks of intent and runtime behavior. We define the language,
19 operational semantics, typing rules, and achievability judgments, and establish
20 refutation soundness, tolerance, capability monotonicity, subject reduction, and
21 session fidelity. The decidable part of the system covers a fixed set of participants;
22 allowing new participants to be added during execution leads to an undecidable
23 extension, for which the checker abstains. In a deterministic evaluation on 15 skills
24 covering common structural failure modes, the checker correctly identifies all 6
25 achievable cases and all 7 genuinely impossible cases. The 2 false *achievable* re-
26 sults are the deliberately constructed cases involving information outside the static
27 guarantee. The complete evaluation takes less than one second on a commodity
28 laptop. The checker itself uses no LLM tokens; in our cost model, compacting
29 the 15 skills costs 12.2K tokens compared with approximately 1.07M tokens for
30 one round of the corresponding agent executions, a roughly 87× reduction. For
31 an achievable skill, the checking cost is only 2.9% of one successful run. These
32 results show that a lightweight formal check can identify structurally impossible
33 skills before an agent spends the time and tokens required for execution.

1 Introduction

A modern agent is no longer merely a chatbot, but an autonomous system capable of taking actions beyond text generation—for example, writing code, sending emails, or executing trades. Such behaviour is often guided by natural-language artifacts that specify instructions, workflows, and constraints for the agent [3, 4, 6, 7]. A *skill* is one common way of expressing such guidance. It is a folder containing instructions and tooling that an agent can discover and load as needed [5], thereby *harnessing* a general-purpose agent toward a particular outcome.

There is growing evidence that this guidance matters. Across seven state-of-the-art open-source multi-agent systems, failure rates range from 41% to 86.7% [8]. More importantly, some of these failures can be addressed without changing the underlying model. In one system, simply adding a task-objective verification step to the specification improves task success by 15.6% [8, Appendix H]. Likewise, repairing the harness based on execution traces improves task completion by 6.3 to 18.4 percentage points across four agent benchmarks [4]. These results show that the instructions and structure surrounding an agent can matter as much as the model itself.

The question, however, is how we can know whether such guidance is sufficient to achieve the outcome it declares. A plan may *book a flight and email a confirmation* even though no email tool is available. It may pursue *a flight under \$500* even though the available booking tool can return only premium fares. Or a planner may wait for a worker’s result while the worker waits for the planner’s answer, with each expecting a message that the other will never send. These are not merely execution failures. In each case, the goal is already unreachable before execution begins.

Such failures are structural and recurring, yet today the only way to discover them is often to run the skill and inspect what went wrong afterwards [4]. By then, the agent may already have spent substantial time and tokens taking actions, and a multi-agent system may already have become stuck waiting for communication that will never occur. A simple static check could, in principle, identify these problems before execution.

This paper asks a deliberately narrow question and answers it with a deterministic formal typing system: **given an LLM skill, can its goal be achieved at all?** We do not, here, verify that every step is safe, nor that the agent behaves correctly on every possible input. Instead, we decide *achievability*. The analysis is tolerant: small details, detours, and payload variation should not trigger a false alarm. At the same time, it is *sound for refutation*: when the compiler concludes that a goal is impossible, that verdict is correct.

1.1 The idea, and the trust boundary

The architecture, shown in Figure 1, has two parts separated by a trust boundary. First, an LLM performs *compaction*: it reads the natural-language skill and turns it into a formal *pack* containing capabilities with preconditions and effects, a goal-marked global protocol, and an initial state. This step is *untrusted*: the LLM may misunderstand the skill or omit relevant details.

A deterministic *checker* then decides achievability using only the resulting pack. The checker is mechanically proved sound: if it returns *impossible*, then the goal is impossible under the formal pack it received. If the LLM omits a constraint or capability from the original skill, the checker cannot recover information that is not present in the pack; it will instead reason about the incomplete pack. Conversely, if the LLM introduces a constraint that was not in the original skill, the checker may report the resulting goal as impossible. Thus, our guarantee is about the checker’s verdict with respect to the formal pack, not about whether the pack perfectly represents the original natural-language skill.

This does not make the compaction step a weakness that we simply ignore. The formal pack provides a structured target for the LLM to extract. Rather than asking the LLM to freely invent a formal model, we give it a predefined logical structure: capabilities have preconditions and effects, goals are explicit, and the protocol follows a fixed syntax. This makes compaction a constrained translation task and gives us a concrete artifact that can be inspected before execution.

Crucially, the gap between *what the user meant* and *what was formalized* cannot be eliminated. We instead make this gap explicit at the compaction step, where the resulting pack can be inspected and checked before execution. When the compaction is accurate, the checker can identify goals that are impossible before the agent spends time and tokens executing the skill. When the compaction is

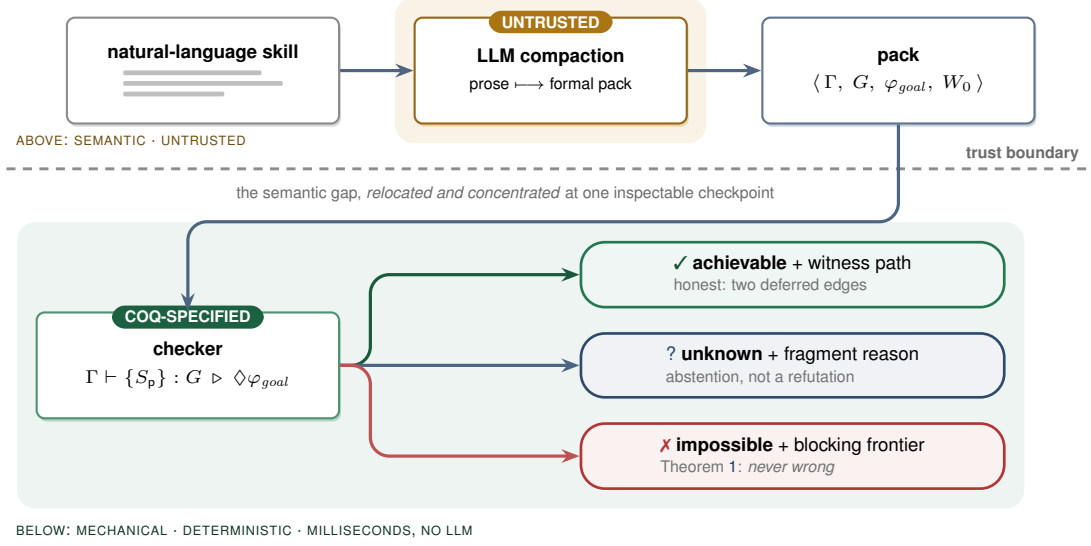


Figure 1: The trust boundary. An untrusted LLM compacts natural-language instructions into a formal pack. A deterministic checker then decides achievability over the pack using a Coq-verified abstraction theorem. The verdicts are asymmetric: *impossible* is sound (Theorem 1), while *achievable* depends on the accuracy of the pack (Section 3.1).

86 inaccurate, the checker still provides a sound verdict for the formal pack it received. The key design
 87 idea is therefore to move achievability checking from runtime execution to an explicit, inspectable
 88 checkpoint before execution.

89 1.2 Contributions

- 90 1. **A goal-achievability type discipline** (Section 3 and Appendices A–C) that combines
 91 capability-guarded reachability from automated planning with deadlock-freedom from
 92 multiparty session types. The discipline checks whole session configurations directly against
 93 a goal-marked global type synthesized from the skill. Building on the multiparty session
 94 type tradition of Honda, Yoshida, and Carbone [12, 13], and the direct session-checking
 95 approach of **SMPS** [1, 2], it adds capabilities, explicit goals, and a refutation-sound but
 96 achievement-incomplete analysis for LLM-drafted skills.
- 97 2. **A mechanized soundness specification** (Appendix D) formalized in Coq. We prove a
 98 refutation-sound abstraction theorem parameterized by simulation hypotheses, together
 99 with tolerance soundness and capability monotonicity: adding capabilities cannot make an
 100 achievable goal impossible. We also establish operational correspondence through subject
 101 reduction and session fidelity for the direct judgment over a labelled transition system. For
 102 single-label communication, we mechanize the cases where other participants continue to
 103 act concurrently; the world-changing action cases are proved on paper under explicit side
 104 conditions.
- 105 3. **A decidability boundary** (Appendix D). Achievability is decidable when the set of par-
 106 ticipants is fixed. We also show that an extension allowing the dynamic addition of an
 107 unbounded number of participants becomes undecidable.
- 108 4. **An implementation and evaluation** (Sections 4–5). We implement the checker together
 109 with an LLM-compaction front-end and a schema gate, and evaluate it on a corpus covering
 110 the main failure modes. The results are consistent with the theoretical guarantees: we
 111 observe no false impossibility verdicts, while every false achievability verdict falls into one
 112 of the two deferred cases.

2 Overview by Example

Consider a one-agent skill whose goal is *a flight is booked and a confirmation is sent*. The LLM compacts the skill into a formal pack. If no email tool is available, there is no action that can establish `confirmation_sent`.

$$\varphi_{goal} = \text{booked} \wedge \text{confirmation_sent}$$

$$\Gamma(a) = \langle pre_a, \langle Add_a, Del_a, Asg_a, Nd_a \rangle \rangle$$

$$\begin{aligned} \text{search} &\mapsto \langle \top, \langle \{\text{searched}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\ \text{filter} &\mapsto \langle \text{searched}, \langle \{\text{filtered}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\ \text{book} &\mapsto \langle \text{filtered}, \langle \{\text{booked}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \\ \text{email} &\mapsto \langle \text{booked}, \langle \{\text{confirmation_sent}\}, \emptyset, \emptyset, \emptyset \rangle \rangle \end{aligned}$$

$$\text{dom}(\Gamma_A) = \{\text{search}, \text{filter}, \text{book}\} \quad \text{dom}(\Gamma_B) = \text{dom}(\Gamma_A) \cup \{\text{email}\}$$

Under STRIPS [17] frame semantics, the checker therefore returns *impossible* and identifies the missing capability. If the email tool is available, the goal is reachable through the sequence `search · filter · book · email`, which the checker returns as a witness. The same example is mechanized in Appendix D (the instance `FlightInstance`). Importantly, the booking itself remains reachable in the first case; the refutation concerns only the missing capability needed to complete the goal.

Tolerance is also important. A skill should not be rejected simply because an agent sends extra status messages, takes a clarifying step, or uses a different payload. Our analysis therefore allows these variations in three ways: *existential may-reachability*, where one successful path is sufficient; *session subtyping*, which allows safe interface variation; and *goal-relevant abstraction*, which ignores payload details that do not affect the goal. These mechanisms are defined in Appendices A–C.

3 Verification Model and Signals

Declared pack. The checker consumes $P = \langle \Gamma, G, \varphi_{goal}, W_0 \rangle$. Here, Γ contains capability declarations with guards and STRIPS-style effects; G is a goal-marked global protocol (a *global type*; Appendix A); φ_{goal} is the logical goal; and W_0 is the initial abstract world state. The LLM produces P from the natural-language skill. A deterministic schema gate enforces the decidable restrictions: finite static roles, guarded recursion, finite-domain booleans, and quantifier-free linear integer arithmetic. The formal syntax and semantics are given in Appendices A–B.

Verification signals. The checker uses three static signals:

Signal	Role	Output
tool availability and effects	capability reachability	missing capability or witness action
coordination structure	global protocol conformance	protocol mismatch or conformant protocol
goal and cost refinements	logical reachability	blocked guard or model

These signals are checked before execution. Runtime payload checks remain separate obligations and are not assumed by the static checker.

Decision procedure. After the schema gate, the checker (i) rejects undeclared capabilities and goal conditions that cannot be achieved by any available capability; (ii) checks role behaviors against G , requiring exact sender choices while allowing safe receiver-side branch supersets; and (iii) explores existential protocol/world successors from W_0 , using Z3 [18] for guards and refinements. A goal condition that cannot be satisfied blocks the search. If the solver times out or the pack falls outside these restrictions, the checker returns `UNKNOWN` rather than *impossible*. The checker therefore returns:

$$\begin{aligned} \text{IMPOSSIBLE}(F) & \quad \text{a sound explanation of why the goal cannot be reached,} \\ \text{ACHIEVABLE}(\pi, O) & \quad \text{a witness path } \pi \text{ and deferred obligations } O, \\ \text{UNKNOWN}(r) & \quad \text{an abstention with a reason } r. \end{aligned}$$

Each verdict records the pack digest, checker semantics, and artifact version.

Guarantees. Let a configuration consist of a protocol state and an abstract world state, and let *abs* map concrete configurations to abstract configurations. Assume step and goal simulation for the declared pack (Appendix D). If no abstract run reaches φ_{goal} , no concrete run represented by that pack reaches the concrete goal (Theorem 1). The analysis remains sound when the abstraction is made less detailed, and adding capabilities cannot turn an achievable goal into an *impossible* one (Theorems 2–3). Direct conformance satisfies subject reduction and session fidelity (Theorems 4–5). The Coq development covers single-label communication with concurrent actions by other participants; world-changing interleavings require explicit stability and commutativity assumptions. The decidable restrictions above guarantee decidability. Allowing an unbounded dynamic addition of participants is an explicit undecidable extension, for which the implementation returns UNKNOWN. Full rules, assumptions, proofs, and mechanization scope appear in Appendix D.

3.1 Structured feedback and human oversight

The checker returns an explanation for IMPOSSIBLE and a witness path for ACHIEVABLE. These outputs provide concrete feedback rather than a single score. A bounded repair adapter may ask the LLM to fix a protocol, but it cannot invent capabilities or weaken the goal. Every revision is checked again by the deterministic checker. Human review remains necessary for the gap between the original intent and the formal pack. Automated prompt or scaffold search is an application of these verdicts, not an empirical contribution of this paper.

4 Implementation

We implemented the checker and its supporting components in Python. The compaction front-end sends the skill to an LLM and passes the result through a deterministic *schema gate* before the checker processes it. Packs that do not satisfy the schema are rejected at this stage. The checker implements the abstractions from Appendices A–C: STRIPS frame semantics for the world, existential search over the protocol, and Z3 for guard and goal satisfiability and arithmetic refinements. On success it returns a witness path; on refutation, it reports the reason, such as a missing capability, unsatisfiable guard, unachievable goal condition, or non-projectable handoff. The optional LLM front-end may use a NON_PROJECTABLE counterexample for one bounded repair round. It may not invent capabilities or weaken the goal, and the deterministic checker remains the final decision maker.

The conformance step ($G \vdash (\mathbb{M}; W)$, Section C.2) is **implemented by means of a simple type inference algorithm since all typing rules are structural in $(\mathbb{M}; W)$. At each step we freely choose either one or two participant processes to reduce or a goal satisfied by the world obtaining the set of sessions which must be typed as premises of the typing rule.**

The adapter also checks the participant side conditions of T-COMM/T-ACT/T-GOAL: it computes $\text{prt}(G)$ and compares it with the roles for which the pack declares behaviour. A declared role outside $\text{prt}(G)$ has contract End, so a non-trivial behavior for that role cannot conform. If the pack declares no behavior for a participant, the checker reports the missing role instead of leaving the assumption implicit.

5 Evaluation

We assembled a corpus of 15 skills covering the main failure modes: hallucinated planning, retry-forever loops, coordination deadlock, and refinement violations. The corpus also includes achievable controls with detours and informed branches, plus two deliberately *spurious* cases for the deferred checks. Each skill has a ground-truth semantic label; the positive class is *achievable*. Evaluation is deterministic and CPU-only: it requires no training or live model inference, completes the 15-case matrix in under one second on a commodity laptop, and gives each individual Z3 query a 10-second timeout.

	predicted achievable	predicted impossible
truly achievable	TP = 6	FN = 0
truly impossible	FP = 2	TN = 7

Two claims, both confirmed. **Soundness:** $FN = 0$ — no achievable goal was wrongly refuted, consistent with Theorem 1 (achievability recall $6/6 = 100\%$). **Incompleteness, contained:** $FP = 2$, and both are the planted spurious cases — one a false payload post-condition (Layer-C obligation), and one an under-specified goal (intent edge). No structural failure was missed in this proof-of-concept corpus. This is evidence of coverage, not a proof that these are the only possible failure modes. Each refutation reason matches its intended failure mode:

Failure mode (source)	Verdict reason
hallucinated planning — invokes a non-existent tool	MISSING_CAPABILITY
hallucinated planning — no establisher for a goal conjunct	GOAL_UNSAT
retry-forever loop — guard never satisfiable	BLOCKED_GUARD
coordination deadlock — unobserved choice / bad handoff	NON_PROJECTABLE
refinement violation — budget unsatisfiable on every run	GOAL_UNSAT

What the check costs. The checker and deterministic front-end use *zero* LLM tokens. The LLM is used once to compact each skill *version*, while a full agent run uses tokens on every *invocation*. Under the conservative model in Appendix E, compacting the 15-spec corpus costs 12.2K tokens and avoids $\sim 1.07M$ tokens per round of invocations, giving $\sim 87\times$ leverage. For every failure mode, the cost of compaction is recovered by the first prevented run. For a skill that is *not* broken, the check costs 2.9% of one successful run, or nothing.

A separate six-case suite tests behavior not covered by the main matrix: tail-recursive retry and non-terminating control, adding new participants during execution, refutation that remains valid when new participants are added, and conformant versus non-conformant declared role behavior. It yields two ACHIEVABLE, three IMPOSSIBLE, and one UNKNOWN. The abstention is reported separately rather than counted as a false negative because UNKNOWN makes no refutation claim. Together the suites contain 21 declared packs; the 15-case matrix remains the only binary ground-truth matrix.

6 Related Work

Multiparty session types [13, 16, 9] provide formal methods for checking communication protocols, including deadlock-freedom through projection and subtyping [14, 10]. Our protocol follows the synchronous formulation of this line of work [9–11]. We also follow the recent approach of checking a whole session configuration directly against a global type [1, 2], rather than first projecting the global type to local types. The new contribution is the world and goal layer: capability guards, effects, goal observation, and the asymmetric refutation guarantee are not modelled by classical MPST or new SMPS. These are not part of classical MPST [13, 16, 9] or the recent direct-checking formulation [1, 2].

Automated planning [17] provides methods for deciding whether a goal can be reached from an initial state using available actions. We use the same basic idea of preconditions and effects, but apply it to agent skills and combine it with multiparty communication protocols. The resulting checker asks whether the declared goal can be achieved given both the available capabilities and the communication protocol.

Recent work on agent verification addresses related problems but has different goals. *Provable coordination via message sequence charts* [20] uses formal communication protocols for LLM agents and focuses on coordination correctness rather than goal achievability. Its runtime-planning extension also studies LLM-generated coordination protocols. *VeriGuard* [21] separates offline policy verification from online monitoring and uses the result as a safety mechanism. *Agent behavioural contracts* [22] provide probabilistic, runtime-enforced compliance and composition conditions for multi-agent systems. In contrast, our checker makes a static achievability decision and guarantees that an *impossible* verdict is correct for the formal pack.

Skill-bundle scanners and verified-skill catalogs address the same deployment object: portable agent skills. NVIDIA’s SkillSpector [23] and Verified Agent Skills pipeline [24] provide checks for skill bundles before deployment, including code, metadata, dependencies, and other security-related properties. These checks are complementary to our approach. They can inspect the original SKILL.md and its supporting files, while our checker operates on the formal pack produced from the

skill. Their purpose is to identify security and consistency problems; ours is to determine whether the declared goal can be achieved with the available capabilities and protocol.

To our knowledge, the work above does not combine capability-based goal reachability with multiparty protocol checking for agent skills, nor provide the same refutation-sound guarantee for an *impossible* verdict.

7 Limitations and Future Work

- **Dependence on the declared pack.** The guarantees apply to the declared Γ and S . If the LLM incorrectly describes a capability or omits an important detail from the skill, the checker can only reason about the resulting pack. A practical system should therefore also inspect `SKILL.md`, scripts, dependencies, manifests, and permission metadata before running the achievability check.
- **Adding participants during execution.** The current procedure is decidable when the set of participants is fixed. When new participants can be added during execution, the procedure may return `UNKNOWN` rather than make an unsupported decision (Proposition 2).
- **Abstraction.** A coarse α allows more behaviors to be treated as equivalent. This preserves the refutation guarantee of Theorem 1, but may make the checker more likely to return *achievable*.
- **Formalization gaps.** The Coq development covers the generic refutation-sound abstraction theorem and concrete examples, head-move action safety, and subject reduction/session fidelity with full bystander interleaving for the single-label communication model, all axiom-free. The executable checker is schema-gated and tested against the formal specification, but its exact symbolic state is not yet fully mechanized. The Coq model also does not yet combine observable goal labels, participant matching, multi-branch choice, and world-changing interleaving in one development. Future work includes completing this mechanization, connecting interleaved session runs to the reachability search, proving equivalence between the declarative judgment and the projection-based adapter, and proving that abstract witnesses correspond to concrete sessions. The 21-pack evaluation is also only a proof of concept.

Broader impact. Pre-execution checking can reduce wasted computation (Appendix E) and prevent structurally impossible plans from reaching execution. However, an incorrect pack can create false confidence, for example by omitting a tool effect or weakening the user’s actual goal. We therefore report the formal pack, the result of the check, and any remaining conditions that still require attention. Human review and runtime checks are still needed for consequential use. This work does not evaluate fairness, privacy, or deployment safety.

8 Conclusion

Agent skills provide instructions, workflows, and tools that guide autonomous agents toward particular outcomes. Whether a skill is written by a human expert or generated with the help of an LLM, a basic question remains: **can the declared goal be achieved at all?** Our work answers this question before execution, using a deterministic checker over a formal pack derived from the skill. The checker combines capability-based reachability with a directly checked global protocol, and its *impossible* verdict is sound for the formal pack it receives.

This guarantee does not remove the gap between the original skill and its formal representation. An incorrect or incomplete pack can lead to an incorrect *achievable* result, which is why the result must be understood relative to the declared pack. Nevertheless, when the pack is correct, the check can identify structural failures before the agent spends time and tokens executing the skill. This provides a simple and inspectable way to check whether an agent skill is achievable before execution.

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333 A Syntax

334 We use disjoint sets of predicates $p \in \text{Pred}$ (boolean), numeric variables $x \in \text{Var}$, roles $p, q, r \in \mathcal{R}$,
 335 capability names $a \in \text{Cap}$, and message labels $\ell \in \text{Lab}$. Following the modern presentation of
 336 multiparty **sessions** [1, 2] we define the model from the bottom up: state and capabilities (Sections A.1–
 337 A.2), then *processes* (Section A.3), and finally the global *type* they are checked against directly
 338 (Section A.4).

339 In each grammar, “ $::=$ ” lists the possible forms of an expression. In each inference rule below,
 340 the statement below the line holds when all premises above the line hold. Rule names identify the
 341 corresponding checker obligation; they are not additional assumptions.

342 A.1 State logic

$$\begin{aligned} e &::= x \mid n \mid e + e \mid e - e \mid c \cdot e & (c, n \in \mathbb{Z}) \\ \varphi &::= p \mid \top \mid \perp \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid e \bowtie e & \bowtie \in \{<, \leq, =, \geq, >, \neq\} \end{aligned}$$

343 The state logic $\mathcal{L}$ is quantifier-free linear integer arithmetic [19] with uninterpreted boolean predicates
 344 — a decidable fragment for which entailment $\models$ and satisfiability are discharged by a satisfiability-
 345 modulo-theories (SMT) solver.

346 A.2 Worlds and capabilities

347 A *world* $W = \langle B, N \rangle$ assigns truth values to predicates $B : \text{Pred} \rightarrow \mathbb{B}$ and numeric variables
 348 $N : \text{Var} \rightarrow \mathbb{Z}$. A *capability context* Γ maps each available tool a to a guarded effect:

$$\begin{aligned} \Gamma(a) &= \langle \text{pre}_a, \text{eff}_a \rangle, & \text{pre}_a &\in \mathcal{L}, \\ \text{eff}_a &= \langle \text{Add}_a \subseteq \text{Pred}, \text{Del}_a \subseteq \text{Pred}, \text{Asg}_a : \text{Var} \rightarrow e, \text{Nd}_a : \text{Var} \rightarrow \mathcal{L} \rangle \end{aligned}$$

349 $a \notin \text{dom}(\Gamma) \iff$ the agent does *not* possess tool a (no hallucinated tools).

350 *Add/Del* are the STRIPS add- and delete-lists; *Asg* gives deterministic numeric updates $x := e$; *Nd*
 351 gives *nondeterministic* updates $x := *$ constrained by a formula over the new value (used to model
 352 post-conditions such as “*books a fare under 500*”).

353 A.3 Processes and multiparty sessions

354 Skills *execute* as processes; a multi-agent run is a session of named processes. We use the synchronous
 355 multiparty session calculus in its modern coinductive **presentation** [1, 2]: **processes coinductively**
 356 **defined are possibly infinite regular trees (finitely many distinct subtrees)**, and labels are goal-relevant
 357 tokens, $\ell = \alpha(m)$ for concrete messages m — the tolerance dial α is a parameter of the label
 358 alphabet.

$$P ::=^{\text{coind}} \mathbf{0} \mid \mathbf{p}!\{\ell_i. P_i\}_{i \in I} \mid \mathbf{p}?\{\ell_i. P_i\}_{i \in I} \mid a. P \quad \mathbb{M} = \mathbf{p}_1[P_1] \parallel \cdots \parallel \mathbf{p}_n[P_n]$$

359 with I finite and non-empty, labels ℓ_i pairwise distinct, and **role** names pairwise distinct. The output
 360 $\mathbf{p}!\{\ell_i. P_i\}$ *internally* chooses a label and sends it to $\mathbf{p}$; the input $\mathbf{p}?\{\ell_i. P_i\}$ offers an *external* choice;
 361 $a. P$ invokes capability a — the only construct that changes the world. A skill S_p is a process: the
 362 behaviour declared for role $\mathbf{p}$.

363 A.4 Global types

364 The single communication construct of a global type is a *directed* labelled choice — selection by $\mathbf{p}$
 365 and branching at $\mathbf{q}$ in one construct. (A partnerless “ $\mathbf{p}$ selects” admits no coherent view for the roles
 366 that did not choose; directed choice is the standard MPST construct [13, 9].)

$$G ::=^{\text{coind}} \mathbf{p} \rightarrow \mathbf{q} : \{\ell_i. G_i\}_{i \in I} \mid a @ \mathbf{p}. G \mid \check{\varphi}. G \mid \text{End}$$

367 subject to $\mathbf{p} \neq \mathbf{q}$ and the same regularity and label conditions as processes. This is the *only* type
 368 in the discipline: Section C.2 types a whole session directly against G — there is no local-type
 369 grammar, no projection function, and no separate subtyping relation. Goal markers $\check{\varphi}$ live *only*
 370 in G : achievability is decided on the global type (Rule ACH in C.3), and markers are consumed at
 371 typing time by T-GOAL without ever burdening a process. Only $a @ \mathbf{p}$ changes the world; messages
 372 carry coordination information only.

B Operational Semantics

B.1 World transitions

A capability fires when its guard holds, producing a successor world. Because Nd is nondeterministic, $\langle W \rangle a \langle W' \rangle$ may relate one world to many. The relation is implicitly indexed by Γ : the notation pre_a, eff_a is defined only when $\Gamma(a) = \langle pre_a, eff_a \rangle$, so every firing presupposes $a \in \text{dom}(\Gamma)$.

$$\frac{\text{WORLD-ACT} \quad W \models pre_a \quad W' \in \text{apply}(eff_a, W)}{\langle W \rangle a \langle W' \rangle}$$

$$\text{apply}(eff_a, \langle B, N \rangle) = \langle B', N' \rangle, \quad B' = (B \cup Add_a) \setminus Del_a$$

$$N'(x) = \begin{cases} \llbracket Asg_a(x) \rrbracket_N & \text{if } x \in \text{dom}(Asg_a) \\ v \text{ with } \langle B, N[x \mapsto v] \rangle \models Nd_a(x) & \text{if } x \in \text{dom}(Nd_a) \\ N(x) & \text{otherwise (frame)} \end{cases}$$

B.2 A labelled semantics for sessions and global types

Only capabilities touch the world, so sessions reduce as configurations $(\mathbb{M}; W)$; communication is synchronous [10]. Each transition carries a label Λ recording its *participants*, so that independent moves by disjoint roles can be identified. Two auxiliary maps drive the labelled system: the *participants* $\text{prt}(\cdot)$ of a global type, a label, or a session, and the *capabilities* $\text{cap}(\cdot)$ (the set of moves a global type can perform). On global types,

$$\begin{aligned} \text{prt}(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}) &= \{p, q\} \cup \bigcup_{i \in I} \text{prt}(G_i) & \text{prt}(\check{\varphi}.G) &= \text{prt}(G) \\ \text{prt}(a @ p.G) &= \{p\} \cup \text{prt}(G) & \text{prt}(\text{End}) &= \emptyset, \end{aligned}$$

on labels $\text{prt}(p \ell q) = \{p, q\}$, $\text{prt}(a @ p) = \{p\}$, $\text{prt}(\check{\varphi}) = \emptyset$, and on sessions $\text{prt}(\mathbb{M}) = \{p \mid \mathbb{M} \equiv p[P] \ \& \ P \neq 0\}$ (the roles with a residual behaviour). The capabilities of a global type are

$$\begin{aligned} \text{cap}(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}) &= \{p \ell_i q \mid i \in I\} \cup \bigcup_{i \in I} \text{cap}(G_i) \\ \text{cap}(a @ p.G) &= \{a @ p\} \cup \text{cap}(G) & \text{cap}(\check{\varphi}.G) &= \{\check{\varphi}\} \cup \text{cap}(G) & \text{cap}(\text{End}) &= \emptyset. \end{aligned}$$

The capability names used by a protocol are

$$\text{caps}(G) = \{a \mid \exists p. a @ p \in \text{cap}(G)\}.$$

The LTS of sessions is defined by

$$\begin{aligned} \text{S-COMM} \quad & \frac{k \in I \subseteq J}{(p[q!\{\ell_i. P_i\}_{i \in I}] \parallel q[p?\{\ell_j. Q_j\}_{j \in J}] \parallel \mathbb{M}; W) \xrightarrow{p \ell_k q} (p[P_k] \parallel q[Q_k] \parallel \mathbb{M}; W)} \\ \text{S-ACT} \quad & \frac{\langle W \rangle a \langle W' \rangle}{(p[a.P] \parallel \mathbb{M}; W) \xrightarrow{a @ p} (p[P] \parallel \mathbb{M}; W')} \\ \text{S-GOAL} \quad & \frac{W \models \varphi}{(\mathbb{M}; W) \xrightarrow{\check{\varphi}} (\mathbb{M}; W)} \end{aligned}$$

where $\parallel$ is commutative, associative and $p[0] \parallel \mathbb{M} \equiv \mathbb{M}$. A non-terminated configuration with no successor is *stuck*; the deadlocking planner/worker handoff of Section 1 is the stuck two-role configuration in which both processes begin with an input. A goal marker is *not* carried by any process; instead S-GOAL lets a configuration *observe* that its world already entails one of its declared goals φ , emitting the label $\check{\varphi}$ without altering $\mathbb{M}$ or W . (Each session carries a set of goal formulas, and φ ranges over that set — not over every formula the world happens to satisfy — so that each observation has a matching marker in the protocol.) This observable goal step is matched at the type level by G-GOAL-E below, so that $\check{\varphi}$ is a genuine label Λ of both transition systems and the operational correspondence of Section D.3 ranges over it uniformly.

The checker relates a session to its protocol through a *labelled* transition system on global types, $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, built from *head* rules that consume the leading construct and *interleaving*

rules that let a move by participants *disjoint* from the head commute inward:

$$\begin{array}{c}
\text{G-COMM-E} \quad \frac{k \in I}{(p \rightarrow q: \{\ell_i.G_i\}_{i \in I}, W) \rightsquigarrow_{p \ell_k q} (G_k, W)} \quad \text{G-ACT-E} \quad \frac{\langle W \rangle a \langle W' \rangle}{(a @ p. G, W) \rightsquigarrow_{a @ p} (G, W')} \\
\\
\text{G-GOAL-E} \quad \frac{W \models \varphi}{(\check{\varphi}. G, W) \rightsquigarrow_{\check{\varphi}} (G, W)} \\
\\
\text{G-COMM-I} \quad \frac{(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W') \quad \Lambda \in \text{cap}(G_i) \quad \forall i \in I \quad \text{prt}(\Lambda) \cap \{p, q\} = \emptyset}{(p \rightarrow q: \{\ell_i.G_i\}_{i \in I}, W) \rightsquigarrow_{\Lambda} (p \rightarrow q: \{\ell_i.G'_i\}_{i \in I}, W')} \\
\\
\text{G-ACT-I} \quad \frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G) \quad \text{prt}(\Lambda) \cap \{p\} = \emptyset}{(a @ p. G, W) \rightsquigarrow_{\Lambda} (a @ p. G', W')} \\
\\
\text{G-GOAL-I} \quad \frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G)}{(\check{\varphi}. G, W) \rightsquigarrow_{\Lambda} (\check{\varphi}. G', W')}
\end{array}$$

395 G-GOAL-E discharges a goal marker when $W \models \varphi$, emitting the observable label $\check{\varphi}$ that matches
396 S-GOAL. In the head-only reduction used for achievability, an unsatisfied marker is a checkpoint and
397 blocks the path. In the full labelled correspondence, G-GOAL-I lets a continuation step commute
398 inward under a still-pending marker. When that step changes the world, the correspondence theorem
399 uses the goal-stability hypothesis stated in Section D.3; removing that hypothesis is future work. The
400 extraction (head) rules G-COMM-E/G-ACT-E/G-GOAL-E give the Λ -erased reduction $(G, W) \rightsquigarrow$
401 (G', W') that the achievability judgment below uses; the interleaving rules matter only for the
402 operational correspondence of Section D.3, and their $\Lambda \in \text{cap}(G)$ premise confines a commuting
403 move to a capability the residual protocol actually offers. G-COMM-E is legitimate because choice
404 is directed — the branch is a communication both endpoints observe, and Section C.2 checks that
405 every third party can follow. A well-formed pack uses its declared goal φ_{goal} at each goal marker. A
406 configuration is *goal-satisfying* when control is at such a marker or at the terminus and the world
407 entails that declared goal:

$$\begin{array}{c}
\text{goal}_{\varphi_{\text{goal}}}(\check{\varphi}_{\text{goal}}.G, W) \iff W \models \varphi_{\text{goal}}, \quad \text{goal}_{\varphi_{\text{goal}}}(\text{End}, W) \iff W \models \varphi_{\text{goal}}, \\
408 \quad \Gamma; G \models \Diamond_{\text{init}} \varphi_{\text{goal}} \iff \exists W_0 \models \text{init}. \exists (G', W'). (G, W_0) \rightsquigarrow^* (G', W') \wedge \text{goal}_{\varphi_{\text{goal}}}(G', W').
\end{array}$$

409 The diamond is *existential*: a single witnessing run suffices — the formal source of detour-tolerance.

410 B.3 Realizability and subtyping, without projection

411 The classical route to ruling out a deadlocking handoff is *projection*: compute each role's local
412 type $G|_r$ by structural recursion on G , taking the *merge* of a choice's branches at every role that
413 neither sends nor receives, and declaring G *realizable* exactly when this partial function is defined
414 everywhere — undefinedness of the merge is the static signature of a role forced to behave differently
415 in branches it cannot distinguish. Session subtyping $\leq$ is then a second, separate coinductive relation
416 on local types, letting a role's actual behaviour offer more external choices and fewer internal ones
417 than its projected contract demands [14, 10, 11].

418 Both devices exist to answer questions about the *network as a whole* — “can every role tell the
419 branches apart it needs to?”, “may this implementation stand in for that contract?” — while type-
420 checking only one role's syntax at a time. Section C.2 answers the same two questions with a single
421 judgment, *typing the whole configuration directly against G*: a choice node's rule quantifies over
422 every branch the protocol declares and requires the *same* residual configuration to check against each
423 one. This is the conservative plain-merge condition, obtained without ever computing $\sqcap$ or a local
424 type. The receiver-side subtyping direction (a receiver may accept a superset of the protocol's labels)
425 is folded into the same rule as a label-set inclusion $I \subseteq J$. Tolerance therefore still has the same
426 three independent knobs from Section 2 — the existential $\Diamond$ (Section B.2), subtyping (now inside

427 T-COMM, Section C.2), and the granularity of α — “flexible, tolerant of small details” is still the
 428 profile $\langle \text{coarse } \alpha, \Diamond, \text{linear } \mathcal{L} \rangle$; a later safety layer flips the same machinery to the universal $\Box$ over
 429 permission predicates.

430 C The Achievability Type System

431 Achievability now factors into three premises, each separately decidable, combined by the top rule
 432 ACH: no hallucinated capability, direct conformance of the declared skills to G , and reachability of
 433 the goal.

434 C.1 Capability and goal typing

$$\begin{array}{c} \text{T-EFF} \\ \frac{\Gamma(a) = \langle pre, eff \rangle \quad \varphi \models pre}{\Gamma \vdash \{\varphi\} a \{\text{sp}(\varphi, eff)\}} \end{array} \quad \begin{array}{c} \text{T-MISS} \\ \frac{a \notin \text{dom}(\Gamma)}{\Gamma \vdash a \triangleright \mathbf{X}(\text{MISSING_CAPABILITY})} \end{array}$$

435 $\text{sp}(\varphi, eff)$ is the strongest postcondition of the STRIPS effect. T-MISS fires on hallucinated planning:
 436 the plan names a tool the context does not grant, and achievability is refuted without any search.

437 C.2 Direct conformance typing

Skills are typed *directly* against the global protocol and the world, by the coinductive judgment $G \vdash (\mathbb{M}; W)$ — read “ G types session $\mathbb{M}$ starting from world W .” There is no local type, no projection, and no separate subtyping relation: the receiver-side subtyping direction and the unobserved-choice check are structural side conditions of one rule.

$$\begin{array}{c} \text{T-END} \\ \frac{}{\text{End} \vdash (\mathbf{p}[0]; W)} \end{array} \quad \begin{array}{c} \text{T-ACT} \\ \frac{G \vdash (\mathbf{p}[P] \parallel \mathbb{M}; W') \quad \langle W \rangle a \langle W' \rangle \quad \text{prt}(G) \cup \{\mathbf{p}\} = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}\}}{a @ \mathbf{p}. G \vdash (\mathbf{p}[a.P] \parallel \mathbb{M}; W)} \end{array}$$

$$\begin{array}{c} \text{T-GOAL} \\ \frac{G \vdash (\mathbb{M}; W) \quad W \models \varphi \quad \text{prt}(G) = \text{prt}(\mathbb{M})}{\checkmark \varphi. G \vdash (\mathbb{M}; W)} \end{array}$$

$$\begin{array}{c} \text{T-COMM} \\ \frac{G_i \vdash (\mathbf{p}[P_i] \parallel \mathbf{q}[Q_i] \parallel \mathbb{M}; W) \quad \forall i \in I \subseteq J \quad \text{prt}(\mathbf{p} \rightarrow \mathbf{q} : \{\ell_i.G_i\}_{i \in I}) = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}, \mathbf{q}\}}{\mathbf{p} \rightarrow \mathbf{q} : \{\ell_i.G_i\}_{i \in I} \vdash (\mathbf{p}[\mathbf{q}!\{\ell_i.P_i\}_{i \in I}] \parallel \mathbf{q}[\mathbf{p}?\{\ell_j.Q_j\}_{j \in J}] \parallel \mathbb{M}; W)} \end{array}$$

438 T-ACT pairs type and world in lockstep with WORLD-ACT: the unconsumed type $a @ \mathbf{p}.G$ sits with
 439 the *pre*-effect world W and the residual G with the *post*-effect world W' — the same convention
 440 as G-ACT-E (Section B.2). In T-COMM the sender $\mathbf{p}$ offers exactly the protocol’s branch set I ,
 441 while the receiver $\mathbf{q}$ may accept *more* ($I \subseteq J$) — the one subtyping direction the rule keeps (safe
 442 interface slack at the receiver). Requiring *every* protocol branch ℓ_i to be checked against the *same*
 443 bystanders $\mathbb{M}$ is the plain-merge form of the unobserved-choice condition, so a role forced to behave
 444 differently across branches it cannot distinguish makes no derivation possible — the rule fails closed,
 445 and deadlock-freedom falls out of T-COMM itself with no merge to compute (Theorem 4). Each
 446 rule additionally carries a *participant-matching* side condition tying the protocol’s participants to
 447 the session’s: $\text{prt}(\mathbf{p} \rightarrow \mathbf{q} : \{\ell_i.G_i\}) = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}, \mathbf{q}\}$ in T-COMM, $\text{prt}(G) \cup \{\mathbf{p}\} = \text{prt}(\mathbb{M}) \cup \{\mathbf{p}\}$
 448 in T-ACT, and $\text{prt}(G) = \text{prt}(\mathbb{M})$ in T-GOAL. These make the invariant “the roles the protocol still
 449 mentions are exactly the roles still running” hold at every node, ruling out a session that carries a stray
 450 participant the protocol has forgotten (or vice versa) — the well-formedness that lets the interleaving
 451 cases of Section D.3 match a bystander move on the session to a $\text{cap}(\cdot)$ -labelled move on the type.
 452 T-END closes the discipline at the terminus: the terminated protocol End types a single idle role $\mathbf{p}[0]$
 453 — and, through the congruence $\mathbf{p}[0] \parallel \mathbb{M} \equiv \mathbb{M}$, any session all of whose roles have terminated —
 454 the base case of the participant invariant, since $\text{prt}(\text{End}) = \emptyset$ and an idle role contributes nothing to
 455 $\text{prt}(\mathbb{M})$. T-ACT is derivable only when $\langle W \rangle a \langle W' \rangle$ holds, which itself presupposes $a \in \text{dom}(\Gamma)$:
 456 this is where hallucinated capabilities die, with T-MISS as the diagnostic.

457 C.3 The achievability judgment

$$\text{ACH} \quad \frac{\text{caps}(G) \subseteq \text{dom}(\Gamma) \quad \exists W_0 \models \text{init}. G \vdash (p_1[S_{p_1}] \parallel \dots \parallel p_n[S_{p_n}]; W_0) \wedge \Gamma; (G, W_0) \models \Diamond \varphi_{\text{goal}}}{\Gamma \vdash \{S_p\}_p : G \triangleright \Diamond \varphi_{\text{goal}}}$$

458 Read top-to-bottom: no hallucinated tools; some plausible initial world exists from which the declared
 459 skills directly conform to the protocol (deadlock-free, every capability call guarded) *and* that same
 460 world reaches the goal. The reachability half is unchanged from Section B.2 and decided on Γ, G
 461 alone, before any S_p is known — exactly what lets the checker run on the LLM-compacted pack
 462 the moment it exists; conformance is the additional, separately decidable check once the skills are
 463 declared, and needs no transport lemma: the receiver-side subtyping slack is already inside T-COMM.

464 C.4 Reachability rules

The diamond is derived by the following inductive system, realized by the checker as a finite forward search over $\rightsquigarrow$; unlike T-COMM/T-ACT/ T-GOAL, it never mentions a session $\mathbb{M}$, which is what makes it decidable on the pack alone.

$$\begin{array}{c} \text{REACH-GOAL} \\ \frac{W \models \varphi_{\text{goal}} \quad G = \text{End} \vee \exists G'. G = \checkmark \varphi_{\text{goal}}. G'}{\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}} \\[10pt] \begin{array}{cc} \text{REACH-STEP} & \text{REACH-OR} \\ \frac{(G, W) \rightsquigarrow (G', W') \quad \Gamma; (G', W') \models \Diamond \varphi_{\text{goal}}}{\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}} & \frac{\exists k \in I. \Gamma; (G_k, W) \models \Diamond \varphi_{\text{goal}}}{\Gamma; (p \rightarrow q: \{\ell_i.G_i\}_{i \in I}, W) \models \Diamond \varphi_{\text{goal}}} \end{array} \end{array}$$

465 D Metatheory

466 We separate what is *mechanized in Coq* (Theorems 1–3; Theorems 4–5 for the single-label com-
 467 munication fragment with full interleaving; the head-move action case; and the concrete instances,
 468 all axiom-free) from what is *proved on paper* (the action-interleaving cases, decidability, and the
 469 undecidability boundary).

470 D.1 Soundness of the abstraction

471 Let $\mathcal{C}$ be the concrete configuration space of protocol/world pairs, $\mathcal{A}$ the abstract configuration space
 472 the checker explores, and $\text{abs} : \mathcal{C} \rightarrow \mathcal{A}$ the abstraction. We require:

$$(\text{step-sim}) \quad C \rightsquigarrow C' \Rightarrow \text{abs}(C) \rightsquigarrow^* \text{abs}(C'), \quad (\text{goal-sim}) \quad \text{goal}_{\varphi_{\text{goal}}}(C) \Rightarrow \hat{\text{goal}}(\text{abs}(C)).$$

473 **Lemma 1** (Transport). *If $C_0 \rightsquigarrow^* C$ then $\text{abs}(C_0) \rightsquigarrow^* \text{abs}(C)$.*

474 *Proof.* By induction on the reflexive-transitive closure. The empty run is reflexivity. For $C_0 \rightsquigarrow^* C'$
 475 $C' \rightsquigarrow C$, the induction hypothesis gives $\text{abs}(C_0) \rightsquigarrow^* \text{abs}(C')$ and STEP-SIM gives $\text{abs}(C') \rightsquigarrow^* \text{abs}(C)$;
 476 compose them. Mechanized as `reach_abs`, parameterized by STEP-SIM. $\square$

477 **Lemma 2** (Reachability monotonicity). *If every step of $\rightsquigarrow_1$ is a step of $\rightsquigarrow_2$ (that is, $x \rightsquigarrow_1 y$ implies
 478 $x \rightsquigarrow_2 y$), then $s \rightsquigarrow_1^* w$ implies $s \rightsquigarrow_2^* w$.*

479 *Proof.* By induction on the closure $s \rightsquigarrow_1^* w$. The empty run transports by reflexivity. For $s \rightsquigarrow_1^* u \rightsquigarrow_1 w$,
 480 the induction hypothesis gives $s \rightsquigarrow_2^* u$; the hypothesis turns the last step $u \rightsquigarrow_1 w$ into $u \rightsquigarrow_2 w$;
 481 appending it gives $s \rightsquigarrow_2^* w$. Mechanized as `reach_mono`. $\square$

482 D.2 Refutation soundness, tolerance, monotonicity

483 **Theorem 1** (Refutation soundness). *If the abstract system has no reachable goal state from
 484 $\text{abs}(G, W_0)$, then no declared run of that pack configuration reaches φ_{goal} . Hence an impossi-
 485 ble verdict is never wrong relative to the pack, provided STEP-SIM and GOAL-SIM hold.*

486 *Proof.* We prove the contrapositive: if *some* declared run reaches the goal, then the abstract
 487 system reaches an abstract goal state. Suppose $C_0 \rightsquigarrow^* C$ with $\text{goal}_{\varphi_{\text{goal}}}(C)$. By Lemma 1,
 488 $\text{abs}(C_0) \rightsquigarrow^* \text{abs}(C)$, and by GOAL-SIM, $\hat{\text{goal}}(\text{abs}(C))$. Thus $\text{abs}(C)$ is an abstract goal state reach-
 489 able from $\text{abs}(C_0)$, contradicting the hypothesis. Hence no declared run reaches φ_{goal} : a verdict
 490 of *impossible* — read off exactly the premise, that the abstract search exhausts without hitting an
 491 abstract goal — is never wrong. Mechanized, axiom-free as a theorem schema parameterized by the
 492 two simulation hypotheses (the existential witness is $\text{abs}(C)$): $\square$

```

493   Theorem refutation_sound : forall s0,
494     (~ exists a, reach astep (abs s0) a /\ agoal a) ->
495     (~ exists w, reach cstep s0 w /\ cgoal w).
496   (* Print Assumptions refutation_sound. => Closed under the global
497     context *)

```

498 **Theorem 2** (Tolerance soundness). *If a coarser over-approximation (more edges) cannot reach the*
 499 *goal, neither can any finer system. Hence coarsening α to tolerate more detail never produces a false*
 500 *refutation.*

501 *Proof.* Let $\rightsquigarrow_{\text{fine}}$ and $\rightsquigarrow_{\text{coarse}}$ be the two abstract step relations, with the coarsening an *over-*
 502 *approximation*: every fine step is also a coarse step, $x \rightsquigarrow_{\text{fine}} y \Rightarrow x \rightsquigarrow_{\text{coarse}} y$. Suppose, for
 503 contradiction, that the finer system reaches the goal: $s_0 \rightsquigarrow_{\text{fine}}^* a$ with $\hat{\text{goal}}(a)$. By Lemma 2 (with
 504 $\rightsquigarrow_1 = \rightsquigarrow_{\text{fine}}$, $\rightsquigarrow_2 = \rightsquigarrow_{\text{coarse}}$), $s_0 \rightsquigarrow_{\text{coarse}}^* a$; and $\hat{\text{goal}}(a)$ still holds. So the coarser system reaches
 505 a goal state, contradicting the hypothesis. Hence no coarser-refuted goal is reachable in any finer
 506 system: coarsening α only *adds* abstract edges, so it can never turn an unreachable goal into a
 507 reachable one — refutation is preserved. Mechanized as `tolerance_sound`, axiom-free. $\square$

508 **Theorem 3** (Capability monotonicity). *If $\Gamma \subseteq \Gamma'$ and Γ reaches φ_{goal} , then Γ' reaches φ_{goal} .*
 509 *Granting tools never makes an achievable goal impossible.*

510 *Proof.* Enlarging the capability context only *adds* guarded effects: since $\Gamma \subseteq \Gamma'$, every capability
 511 firing available under Γ is available under Γ' with the same guard and effect (WORLD-ACT in B.1
 512 quantifies over $a \in \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$), so every step of the Γ -transition relation is a step of
 513 the Γ' -relation. Suppose Γ reaches the goal configuration C from C_0 . By Lemma 2, the same
 514 configuration is reachable under Γ' , and $\text{goal}_{\varphi_{\text{goal}}}(C)$ is unchanged because the goal predicate does
 515 not depend on Γ . Hence Γ' reaches φ_{goal} via the same witnessing run. The other premises of ACH
 516 are monotone in the same direction ($\text{caps}(G) \subseteq \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$ still holds, and realizability
 517 and conformance do not mention Γ beyond $a \in \text{dom}(\Gamma)$), so the whole judgment is preserved. The
 518 transport half is mechanized as `cap_monotone`, axiom-free, stated over an inclusion of step relations;
 519 the reduction of $\Gamma \subseteq \Gamma'$ to that inclusion is the WORLD-ACT argument above, on paper. $\square$

520 **Concrete instance.** The flight skill of Section 2, variant A, is mechanized as `FlightInstance`. We
 521 prove `flight_refuted` (no run reaches `booked` $\wedge$ `confirmation_sent`, since no step touches `conf`)
 522 and `booking_reachable` (a booking is still reachable). Both are axiom-free, witnessing non-vacuity
 523 of Theorem 1.

524 D.3 Operational correspondence: subject reduction and session fidelity

525 The direct judgment $G \vdash (\mathbb{M}; W)$ is not merely preserved by reduction: the session and its protocol
 526 step in *lockstep* over the labelled transition systems of Section B.2. This is the standard soundness
 527 backbone of a session calculus [13, 11], obtained here *directly*, with no projection and no merge. For
 528 world-changing interleavings, the statements below assume two explicit frame conditions: effects of
 529 participant-disjoint actions commute, and an action moved under a pending marker preserves that
 530 marker's formula. Communication-only reductions satisfy both conditions vacuously.

531 **Lemma 3** (Residual capability). *If $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, then $\Lambda \in \text{cap}(G)$.*

532 **Lemma 4** (Participant agreement). *If $G \vdash (\mathbb{M}; W)$, then $\text{prt}(G) = \text{prt}(\mathbb{M})$.*

533 Both follow by induction on the transition or typing derivation, respectively; they are the capability-
 534 membership and participant invariants used in the interleaving cases below.

535 **Theorem 4** (Subject reduction). *If $G \vdash (\mathbb{M}; W)$ and $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$, then $(G, W) \rightsquigarrow_{\Lambda}$*
 536 *(G', W') for some G' with $G' \vdash (\mathbb{M}'; W')$.*

537 **Theorem 5** (Session fidelity). *If $G \vdash (\mathbb{M}; W)$ and $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, then $(\mathbb{M}, W) \xrightarrow{\Lambda}$*
 538 *$(\mathbb{M}', W')$ for some $\mathbb{M}'$ **and** W' with $G' \vdash (\mathbb{M}'; W')$.*

539 We prove both by induction on the typing derivation $G \vdash (\mathbb{M}; W)$, with a case analysis on the
 540 transition; the one auxiliary fact used throughout is that typing respects pointwise-equal sessions
 541 (`ctypes_ext`), which lets the interleaving cases reassociate independent updates.

542 *Proof of Theorem 4 (subject reduction).* By induction on $G \vdash (\mathbb{M}; W)$.

543 *Case T-END* ($G = \text{End}$, every role idle). Then $\mathbb{M}$ has no output or pending action, so the only
 544 available transition is the goal observation S-GOAL at $\Lambda = \check{\varphi}_{goal}$ when $W \models \varphi_{goal}$, which leaves
 545 $(\mathbb{M}, W)$ fixed; there is no residual type to preserve and the claim holds (otherwise no Λ -transition
 546 exists and it holds vacuously).

547 *Case T-GOAL* ($G = \check{\varphi}.G_0$, with $W \models \varphi$ and $G_0 \vdash (\mathbb{M}; W)$, $\text{prt}(G_0) = \text{prt}(\mathbb{M})$). The transition
 548 $(\mathbb{M}, W) \rightarrow_{\Lambda} (\mathbb{M}', W')$ splits on its label.

- 549 • *Goal observation* ($\Lambda = \check{\varphi}$, by S-GOAL, so $\mathbb{M}' = \mathbb{M}$ and $W' = W$). Since $W \models \varphi$, rule
 550 G-GOAL-E discharges the marker, $(\check{\varphi}.G_0, W) \rightsquigarrow_{\check{\varphi}} (G_0, W)$, and $G_0 \vdash (\mathbb{M}; W)$ is the
 551 premise: take $G' = G_0$.
- 552 • *Interleaving* ($\Lambda \neq \check{\varphi}$). A goal marker is not a session construct, so this is already a
 553 transition of G_0 's session; the induction hypothesis gives $(G_0, W) \rightsquigarrow_{\Lambda} (G', W')$ with
 554 $\Lambda \in \text{cap}(G_0)$ and $G' \vdash (\mathbb{M}'; W')$. When the move is world-preserving (a communica-
 555 tion, $W' = W$), G-GOAL-I carries the still-pending marker along, $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda}$
 556 $(\check{\varphi}.G', W)$, and $\check{\varphi}.G' \vdash (\mathbb{M}'; W)$ by T-GOAL (the side condition $\text{prt}(G') = \text{prt}(\mathbb{M}')$ is
 557 preserved because Λ acts on the same participants on both sides). When the move is *world-*
 558 *changing* (a capability firing, $W' \neq W$), G-GOAL-I still commutes it under the marker,
 559 $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda} (\check{\varphi}.G', W')$; re-typing the residual by T-GOAL then additionally asks
 560 $W' \models \varphi$. This follows from the goal-stability hypothesis on actions commuting under pend-
 561 ing markers. The global type sequences the marker before G_0 's actions whereas the session
 562 may fire the action first; the interleaving rule lets the type catch up under that hypothesis.
 563 Removing the hypothesis, rather than this conditional case, is the open correspondence
 564 problem recorded in Section 7.

565 *Case T-COMM*, with head $p \rightarrow q$ and branches G_i . Here $\mathbb{M}$ is $p[q! \dots] \parallel q[p? \dots] \parallel \mathbb{M}_0$, and each
 566 branch satisfies $G_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0; W)$. If the transition is a goal observation $\Lambda = \check{\varphi}$
 567 (S-GOAL, leaving $(\mathbb{M}, W)$ fixed), it is matched by G-COMM-I: $\text{prt}(\check{\varphi}) = \emptyset$ makes the disjointness
 568 premise trivial, and each branch realises the same observation by the induction hypothesis, using
 569 $\check{\varphi} \in \text{cap}(G_i)$ — the observation concerns a marker the protocol actually pends (see the note below).
 570 Otherwise the transition is a communication $\Lambda = r \ell t$. Because p offers only outputs to q and q only
 571 inputs from p , the sender/receiver shapes force a dichotomy.

- 572 • *Head move* ($r = p$). Then $t = q$ and $\ell = \ell_k \in I$, and $\mathbb{M}' = p[P_k] \parallel q[Q_k] \parallel \mathbb{M}_0$. Rule
 573 G-COMM-E gives $(G, W) \rightsquigarrow_{p\ell_k q} (G_k, W)$, and $G_k \vdash (\mathbb{M}'; W)$ is exactly the k -th premise.
- 574 • *Interleaving move* ($r \neq p$). The shapes then force $r, t \notin \{p, q\}$: r is an output so $r \neq q$,
 575 and t is an input so $t \neq p$, while $t = q$ would force $r = p$. Since r, t lie in the bystanders
 576 $\mathbb{M}_0$, the *same* communication is enabled in each branch's configuration $p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0$
 577 (the head updates do not touch r, t). The induction hypothesis at each i yields G'_i with
 578 $(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W)$ and $G'_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}'_0; W)$, where $\mathbb{M}'_0$ is $\mathbb{M}_0$ after the
 579 bystander step. As $\text{prt}(\Lambda) = \{r, t\}$ is disjoint from $\{p, q\}$, rule G-COMM-I assembles
 580 $(G, W) \rightsquigarrow_{\Lambda} (p \rightarrow q: \{\ell_i, G'_i\}, W)$, and re-applying T-COMM (the sender/receiver at p, q
 581 are untouched, and each branch is retyped by the induction hypothesis, reassociating the
 582 independent updates via `ctypes_ext`) gives the residual typing.

583 Communications leave the world fixed, so no independence hypothesis is needed. For a world-
 584 changing action at the head (T-ACT), G-ACT-E matches and the residual G is typed at the *same*

585 post-effect world the rule reaches — this is exactly where the lockstep world pairing of the corrected
 586 T-ACT is used; a goal observation at an action-typed configuration is matched by G-ACT-I exactly
 587 as in the T-COMM case, since $\text{prt}(\checkmark\varphi) \cap \{p\} = \emptyset$ trivially. Interleaving *two* actions requires their
 588 effects to commute ($\text{eff}_a \circ \text{eff}_b = \text{eff}_b \circ \text{eff}_a$ for disjoint a, b), a frame-independence side condition
 589 that participant-disjointness alone does not supply; under it the T-ACT interleaving case goes through
 590 identically. $\square$

591 *Proof of Theorem 5 (session fidelity).* By induction on $G \vdash (\mathbb{M}; W)$, inverting the global step
 592 $(G, W) \rightsquigarrow_{\Lambda} (G', W')$. T-END admits only the goal observation G-GOAL-E at $\checkmark\varphi_{\text{goal}}$, matched
 593 on the session by S-GOAL. For T-GOAL, the step is either G-GOAL-E — matched by S-GOAL
 594 on the session, the marker discharged — or G-GOAL-I, whose premise $(G_0, W) \rightsquigarrow_{\Lambda} (G'_0, W')$
 595 the induction hypothesis realises as an underlying session step that carries the marker along. For
 596 T-COMM, the global step is either G-COMM-E — and the prescribed head communication $p \ell_k q$ is
 597 enabled by S-COMM, landing in the k -th premise — or G-COMM-I with $\text{prt}(\Lambda) \cap \{p, q\} = \emptyset$; then
 598 the induction hypothesis on any branch produces a bystander communication, whose endpoints lie
 599 outside $\{p, q\}$, so S-COMM fires it on $\mathbb{M}$ and T-COMM retypes the result. The action cases mirror
 600 those of Theorem 4. $\square$

601 **Mechanization.** Both theorems are mechanized axiom-free for the single-label communication
 602 fragment — the fragment in which bystander interleaving is unconditional — in `DirectTypingSR.v`,
 603 over labelled session and global transition systems, with no projection, merge, or local type:

```
604   Theorem subject_reduction : forall W G s s' L,
605     ctypes W G s -> lstep L s s' ->
606     exists G', gstep W L G G' /\ ctypes W G' s'.
607   Theorem session_fidelity : forall W G G' s L,
608     ctypes W G s -> gstep W L G G' ->
609     exists s', lstep L s s' /\ ctypes W G' s'.
610   (* Print Assumptions => Closed under the global context (both). *)
```

611 The head-move case for world-changing *actions* is additionally mechanized in `DirectTyping.v`
 612 (`type_directed_safety`, also `progress`). Observable goal labels appear in neither Coq develop-
 613 ment: the mechanized label alphabet is communication-only, and its goal rule fuses marker discharge
 614 with a continuation step. Participant-matching side conditions, labelled action interleaving, and
 615 multi-branch choice are likewise absent. Completing a faithful mechanization of the paper rules
 616 remains work in Section 7.

617 **Concrete instance.** `HandoffInstance` mechanizes the planner/worker handoff of Section 1 on both
 618 sides. The *good* pairing — planner sends, worker delivers, goal checked — is proved `ctypes`-typed
 619 (`good_typed`) and its run is proved to reach a world satisfying the goal (`good_runs_to_goal`). The
 620 *bad* pairing — both roles start with an input, exactly the deadlock of Section 1 — is proved `Stuck`
 621 (`bad_stuck`), and the contrapositive of `progress` then gives, for free, that *no non-trivial global type*
 622 *can ever type it* (`bad_untypeable`): the rule rejects the classical deadlocking handoff without any
 623 projection ever being computed.

624 D.4 Incompleteness, by design

625 **Proposition 1** (Incompleteness). $\Gamma; G \models \Diamond_{\text{init}} \varphi_{\text{goal}}$ may hold while no concrete runtime execution
 626 reaches φ_{goal} : the witnessing abstract path can be spurious when α collapses a distinction that
 627 mattered. The system is sound for $\times$ and incomplete for $\checkmark$.

628 This is the price of decidable, tolerant over-approximation; tightening α trades tolerance for precision.
 629 The residue is confined to two edges, motivating the layered architecture:

- 630 • **Payload faithfulness** (bottom edge): a tool’s declared post-condition (“filters to fares
 631 < 500”) may be false — a runtime obligation for a deterministic monitor (Layer C), not a
 632 static one.
- 633 • **Intent fidelity** (top edge): the formalized goal may not capture what the user meant —
 634 irreducibly semantic, surfaced at the single compaction checkpoint for human review.

635 D.5 Decidability and the autonomy boundary

636 **Theorem 6** (Decidability). *In the fragment with finitely many roles (static topology), regular global*
 637 *types represented by finite tail-recursive control graphs, decidable state logic $\mathcal{L}$, and $\mathcal{L}$ -definable*
 638 *effects, $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ is decidable.*

639 *Proof.* The pack-level achievability judgment $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ (Section B.2) is an existential
 640 reachability query over configurations (G', W) ; we exhibit a terminating decision procedure and
 641 argue its completeness.

642 *The search space is finite.* A configuration has two components. (i) *Control*, the residual global
 643 type G' reachable from G by $\rightsquigarrow$. Since G is a regular tree (tail-recursion: finitely many distinct
 644 subtrees, Section A.3) and each $\rightsquigarrow$ step either descends into a subtree (G-COMM-E, G-ACT-E) or
 645 erases a marker (G-GOAL-E), the set $\{G' : (G, _) \rightsquigarrow^* (G', _)\}$ is a subset of the finitely many
 646 subtrees of G , hence finite. (ii) *Symbolic world*. The boolean part is a valuation $B : \text{Pred} \rightarrow \mathbb{B}$
 647 over the finitely many predicates named in the pack, so there are $2^{|\text{Pred}|}$ boolean states. The numeric
 648 part is summarized by an accumulated constraint $\psi \in \mathcal{L}$ (a conjunction of the guards taken and
 649 the Nd -postconditions assumed); on a control-graph back edge the reference procedure applies the
 650 widening $\psi \mapsto \top$, forgetting accumulated numeric constraints. This one-element numeric summary
 651 is an over-approximation and, together with the finite boolean valuations, leaves only finitely many
 652 symbolic worlds. The product with finite control is therefore finite.

653 *Each edge is computable.* Firing a capability requires checking $W \models pre_a$ and forming sp ; testing
 654 goal-satisfaction requires $W \models \varphi$. Both are entailment/satisfiability queries in quantifier-free linear
 655 integer arithmetic with uninterpreted predicates — a decidable theory, discharged by an SMT solver.
 656 Branch selection (REACH-OR) is a finite disjunction. Existence of an initial world is decided by the
 657 same SMT query over $init$; each satisfying symbolic initial valuation seeds the finite search.

658 *Termination and correctness.* Forward search (breadth-first over $\rightsquigarrow$) maintains a visited set of (control,
 659 symbolic-world) pairs; because that set is finite it adds each pair at most once, so the search halts. It
 660 answers ACHIEVABLE iff it enumerates a goal-satisfying configuration, and IMPOSSIBLE otherwise.
 661 Soundness of the IMPOSSIBLE verdict is Theorem 1; and widening only *enlarges* the reachable set (it
 662 weakens constraints, adding edges), so by Theorem 2 it never introduces a false refutation. Hence the
 663 procedure is a decision procedure for $\Gamma; G \models \Diamond_{init} \varphi_{goal}$. $\square$

664 **Proposition 2** (Undecidability in a dynamic-topology extension). *Consider the extension G^+ of the*
 665 *grammar with spawn $r.G$, which creates fresh participants at run time. With an unbounded role*
 666 *population and dynamic channels, achievability in G^+ is undecidable.*

667 *Proof sketch.* By reduction from the control-state reachability problem for communicating finite-
 668 state machines (CFSMs), which is undecidable [15]. Fix a CFSM instance: finite-control machines
 669 M_1, M_2 exchanging messages over unbounded FIFO channels, and a target control state q_f of M_1 ;
 670 deciding whether a run reaches q_f is undecidable.

671 We compute, from this instance, a skill whose goal is achievable iff the CFSM reaches q_f . Each
 672 machine M_i becomes a role whose process mirrors its control graph: a transition that sends message
 673 m spawns, at run time, a fresh *courier* participant carrying m (this is exactly the dynamic-spawning
 674 capability the theorem hypothesizes), and a transition that receives m synchronizes with the oldest
 675 outstanding courier for that channel, so the unbounded family of couriers realizes the unbounded FIFO.
 676 The required order can be made explicit by assigning sequence numbers through the numeric world
 677 state and guarding receives on the next sequence number; the *Asg/Nd* machinery of Section A.2
 678 supplies those updates. A boolean predicate at_{q_f} is established by the single capability guarding
 679 M_1 's entry into q_f , and we set $\varphi_{goal} = at_{q_f}$. By construction a run of the skill reaches a state
 680 satisfying φ_{goal} iff the CFSM has a run reaching q_f : the courier discipline then puts the skill's
 681 reachable configurations in correspondence with the CFSM's channel contents and control states.
 682 The construction is a computable map from CFSM instances to packs, so a decision procedure for
 683 achievability would decide CFSM control-state reachability — impossible. Hence achievability is
 684 undecidable once participants may be spawned dynamically. (The finiteness argument of Theorem 6
 685 breaks at exactly the point this reduction exploits: an unbounded number of couriers makes the set of
 686 reachable controls infinite.) $\square$

687 The two results isolate one concrete boundary: static finite topology admits a total decision procedure,
688 while unbounded dynamic spawning does not. This is a boundary on topology, not a claim that
689 all forms of agent autonomy are undecidable. The implementation returns UNKNOWN when a
690 pack violates the static-topology side condition of Theorem 6, unless an independent capability or
691 establisher refutation has already proved it impossible.

692 D.6 The partition of obligations

693 **Corollary 1.** *The architecture separates “will the goal be achieved” into four obligations on disjoint*
694 *substrates:*

Concern	Where	Mechanism
goal achievability	static, decidable	Coq specification; deterministic search; no LLM
per-step safety / least privilege	Layer B (future)	same engine, $\square$ + permission tiers
payload faithfulness	Layer C (future), run-time	deterministic monitors with blame
intent fidelity	top edge, synthesis	one human checkpoint

696 D.7 Verifiable feedback beyond a scalar reward

697 The verdict is a heterogeneous verification signal rather than a scalar score. An IMPOSSIBLE result
698 identifies which premise failed and carries a blocking frontier; ACHIEVABLE carries a witness
699 path and explicitly retains the two deferred obligations; UNKNOWN is an abstention outside the
700 decidable fragment, not a refutation. These structured outcomes can serve as cheap negative labels
701 for automated skill or prompt search without an LLM judge, while the intent-fidelity checkpoint
702 keeps the irreducible semantic decision with a human. We treat such search as an application of the
703 verifier, not as a contribution evaluated here.

704 E Token economics of pre-execution refutation

705 The practical case for deciding achievability *before* execution is an economic one, so we state it in
706 tokens rather than adjectives. Two asymmetries carry it. First, no model sits in the trusted path: the
707 schema gate, projection, conformance and Z3 reachability spend **zero tokens**, and the deterministic
708 front-end spends zero as well. Only the optional LLM compaction spends tokens, and it is paid *once*
709 *per skill version*. Second, an agent turn re-sends its whole context: writing S for the harness prompt,
710 K for the loaded skill, g for the mean tokens appended per turn and o for output per turn, a run of T
711 turns reads $T(S+K) + gT(T-1)/2$ input and To output tokens. Compaction is linear in the skill
712 and paid once; a doomed run is *quadratic* in its turns and recurs on every invocation.

713 Runtime waste is necessarily a model — it prices a run that, if the refutation is correct, never
714 happens — so we publish the model rather than a single number. Each refutation reason carries
715 a (low, typical, high) band for how long an agent flails before that failure stops it, and how many
716 agents are billed. With conservative defaults ($S=2500$, $g=700$, $o=250$) and a median real consumer
717 skill ($K \approx 1.7K$ tokens), compaction costs $\sim 2.8K$ tokens against the following avoided waste per
718 prevented invocation:

Verdict reason	turns (lo–typ–hi)	waste/run	leverage
MISSING_CAPABILITY	3–8–14, 1 agent	50 240	18×
GOAL_UNSAT	8–18–30, 1 agent	176 040	63×
NON_CONFORMANT	6–15–30, 2 agents	261 900	94×
NON_PROJECTABLE	6–15–30, 2 agents	261 900	94×
BLOCKED_GUARD	12–25–50, 1 agent	305 750	110×

720 The ordering is the robust part, and it is the expected one. MISSING_CAPABILITY is cheapest at
721 run time: the agent calls a tool that is not there and learns immediately, and most of the cost is
722 the improvisation that follows. BLOCKED_GUARD is dearest, because a precondition no run can
723 satisfy is precisely the case an agent cannot learn from — it retries to the harness turn cap. The
724 two multi-party reasons bill two contexts. GOAL_UNSAT additionally carries a cost the token bill

725 does not show: a run whose goal conjunct has no establisher can terminate *believing* it succeeded.
 726 DYNAMIC_TOPOLOGY claims no savings at all, because an abstention prevents nothing.

727 Over the 15-spec corpus the seven structural refutations cost 12.2K tokens of compaction and avoid
 728 $\sim 1.07\text{M}$ tokens per round of invocations (band 0.28M–3.15M): $\sim 87\times$ leverage, or nothing at all
 729 on the deterministic path. Break-even therefore falls *inside* the first prevented run in every mode
 730 — between 0.9% and 5.5% of it. The honest denominator for a healthy skill, where the check buys
 731 nothing and still costs something, is 2.9% of one successful run over the corpus and 4.0% for a
 732 median real skill; with the deterministic front-end, zero. Prompt caching changes the price of the
 733 wasted tokens, not their number. Halving every waste default leaves leverage at $9\times$ – $55\times$ and the
 734 ordering unchanged. The model, its defaults and their justification are in the artifact (`skillc.tokens`,
 735 `skillc.cost`); the Coq development costs no tokens and is amortized over every skill ever checked.

736 NeurIPS Paper Checklist

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738 Question: Do the main claims made in the abstract and introduction accurately reflect the
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740 Answer: [\[Yes\]](#)

741 Justification: The abstract and Section 1 state that refutation is sound relative to the declared
 742 pack, distinguish UNKNOWN from refutation, and identify the mechanized and paper-only
 743 proof fragments.

744 2. Limitations

745 Question: Does the paper discuss the limitations of the work performed by the authors?

746 Answer: [\[Yes\]](#)

747 Justification: The Limitations and Future Work section discusses relative soundness, static
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749 3. Theory assumptions and proofs

750 Question: For each theoretical result, does the paper provide the full set of assumptions and
 751 a complete (and correct) proof?

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 758 perimental results of the paper to the extent that it affects the main claims and/or conclusions
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840 Answer: [N/A]

841 Justification: The paper’s core method does not depend on an LLM. The main contributions—
842 the achievability type discipline, its formal metatheory, and the deterministic checker—are
843 defined and verified independently of LLM inference. The LLM is used only as a front-end
844 to compact a natural-language skill into the formal pack consumed by the checker; the
845 same formal pack can be provided directly without an LLM. The formal guarantees and the
846 reported evaluation therefore do not depend on the behavior of any particular LLM.